DIY information provision with mobilityHubTools: CERC presentation notes

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- *Situation:* Shared mobility hubs[1] are a (relatively) new direction in transport. They seek to co-locate shared, micro-mobility transport modes, sometimes with electrified and/or public transport services or other facilities that might help the movement of people and goods.[2] These tend to aim towards providing more options for those, more occasional, trips that are difficult to make or otherwise impractical on foot, bicycle or using transit, and so contribute to more sustainable travel behaviors.[3]
- *Problem:* Not much is yet known about providing users with information at mobility hubs. While there is some guidance from CoMoUK and others, as of yet there do not appear to be tools available to support practitioners implementing hubs. Rather, mobility hub information provision appears to be largely done on a site-by-site, system-by-system or otherwise bespoke.
- *Question:* Is it possible to improve understandings of current (and best) practices, and to develop tools to fill this gap?
- *Answer:* Undertake design and case-based research guided by the iterative Double Diamond design process to examine information at selected mobility hubs, and to discover current practices. Then define what mobility hub information provision might be trying to solve, develop design solutions and deliver a package of software.

Background

Draft intro video
https://www.youtube.com/watch?v=hu9jsgi8CVU

Draft mobility hub website
https://james-reynolds.github.io/ROBUST_Electric_shaRed_mOBility_hUbS_Trial_website/

Draft mobility hub website, Github source code
https://github.com/James-Reynolds/ROBUST_Electric_shaRed_mOBility_hUbS_Trial_website

Figure 1: Draft mobility hub video and website links

[1] "A shared mobility hub is a place where different transport modes are integrated seamlessly, promoting efficient and sustainable urban mobility. Emphasis is given on shared mobility options such as bikes, scooters and cars. Connection with public transport (transit) is desirable but not necessary. The integration between mobility suppliers is important to provide a seamless, flexible connection at these transfer points. Ensuring an enjoyable experience for travelers is a crucial part of the hub concept as well, and therefore the hub should be considered not just as a transfer node but as a place where a variety of services come together, such as logistics services or restaurants." (Roukouni et al. 2023)

[2] Anastasia Roukouni et al., "An Analysis of the Emerging 'Shared Mobility Hub' Concept in European Cities: Definition and a Proposed Typology," Article, *Sustainability (Switzerland)* 15, no. 6 (2023), doi:10.3390/su15065222.

[3] Karst Geurs et al., "The Smarthubs Integration Ladder: A Conceptual Model for the Categorisation of Shared Mobility Hubs," *Transport Reviews* 44, no. 1 (2024): 112–39, doi:10.1080/01441647.2023.2239499.

This is part of the Electric shaRed mOBility hUbs Trial (ROBUST) project[4]. The project includes co-design, behaviour nudging and hub incubation activities (Figure 1).

[4] See https://www.tcd.ie/transport-research/our-research/research-projects/ROBUST-electric-shared-mobility-hubs-trial/

The literature suggests: branding, signage, guidance, 'totem' displays, and online information CoMoUK;[5] and notes a need for: "integration", "seamless, flexible connection" and ". . . an enjoyable experience for travelers" (Roukouni et al. 2023).

[5] *Communications: How to Increase Awareness and Support for Mobility Hubs*, 2021, https://tinyurl.com/5cxuxeru.

Figure 2: CoMoUK increasing awareness case study

Methodology

- four phases,
- alternate between divergent (discover, develop) and convergent (define, deliver) approaches,
- overall process is iterative,
- industrial design research approach, with broad applications to transport.[6]

Figure 3: Framework for Innovation. Source : UK Design Council (undated)

[6] Selby Coxon, Robbie Napper, and Mark Richardson, *Urban Mobility Design* (Elsevier, 2018), doi:10.1016/C2017-0-01129-1.

Discovery and definition: case-based

- effective when research at an early stage, and for 'how' or 'why' questions (How is information provided at and about mobility hubs? How might information at and about mobility hubs be provided ideally? How can such best practices can be made generally applicable?).

- select cases for theoretical reasons (e.g. leading, notable or representative, or for polar opposites for cross-case comparison etc.)(Robert Yin[7] and others)[8].
- duality criterion: at the same time grounded within the context of the small number of cases (depth of research), yet seeking generalisable results that apply to more than just the cases studied (Martyn Denscombe[9] and others).

DEVELOPMENT AND DELIVERY: iterative coding to replicate various examples; develop synthesis version (industry practice + respond to defined problem); and deliver as a software package.[10]

Results

Discovery: wide range of examples[11].

Definition of the problem to be addressed by the design output: 1. What is this? 2. Where am I? 3. What is here? 4. What is nearby? 5. How can I travel from here? (6. How do I use each bit of this?)[12].

Development of: Bristol-style totem, Birmingham-style totem and 'synthesis' totem. Examples in Figure 5, but also up on the walls around us.

Delivery of: R package on github[13]; code developed to produce paper outputs[14]; and these notes and the associated outputs used in this presentation[15].

Discussion and conclusions

- Tactical urbanism: why not make a "mobility hub" of your own?
- More to do to improve the code, develop further outputs etc.
- Open Street Maps data quite powerful, as are the GIS tools available for R.

[7] *Case Study Research and Applications: Design and Methods*, Sixth (Thousand Oaks, CA, USA: SAGE Publications, Inc., 2018).

[8] Cases: (*Redbridge*) South Woodford and High Street, Wanstead; (*Camden*) Goldington Crescent, and Doric Way; (*Bristol*) Abbey Wood, UWE Frenchay, and Gainsborough Square; and (*Birmingham*) Huntingtree Park and Andrew Road

[9] *The Good Research Guide*, 2nd ed. (Maidenhead, Berkshire, UK: McGraw-Hill International UK Ltd, 2007).

[10] Hadley Wickham and Jennifer Bryan, *R Packages* (" O'Reilly Media, Inc.", 2023), `https://r-pkgs.org/`.

[11] `https://maps.app.goo.gl/1dd8sEJCiwCgPkc98`, `https://maps.app.goo.gl/j7b7MLr8HzcjqoS66`, `https://maps.app.goo.gl/BW8cEVXioYSoNbiz9`, `https://maps.app.goo.gl/1JqvPLWkwkbnyGB27`

Figure 4: Selected hub images. Source: authors.

[12] This is coming out of subsequent co-design workshops.

[13] `https://github.com/James-Reynolds/mobilityHubTools`

[14] `https://github.com/James-Reynolds/WP5_mobilityHubTools_papers/blob/main/ReynoldsEtAl2026_CERC.Rmd`

[15] `https://github.com/James-Reynolds/WP5_mobilityHubTools_papers/blob/main/ReynoldsEtAl2026_CERC_preso.Rmd`

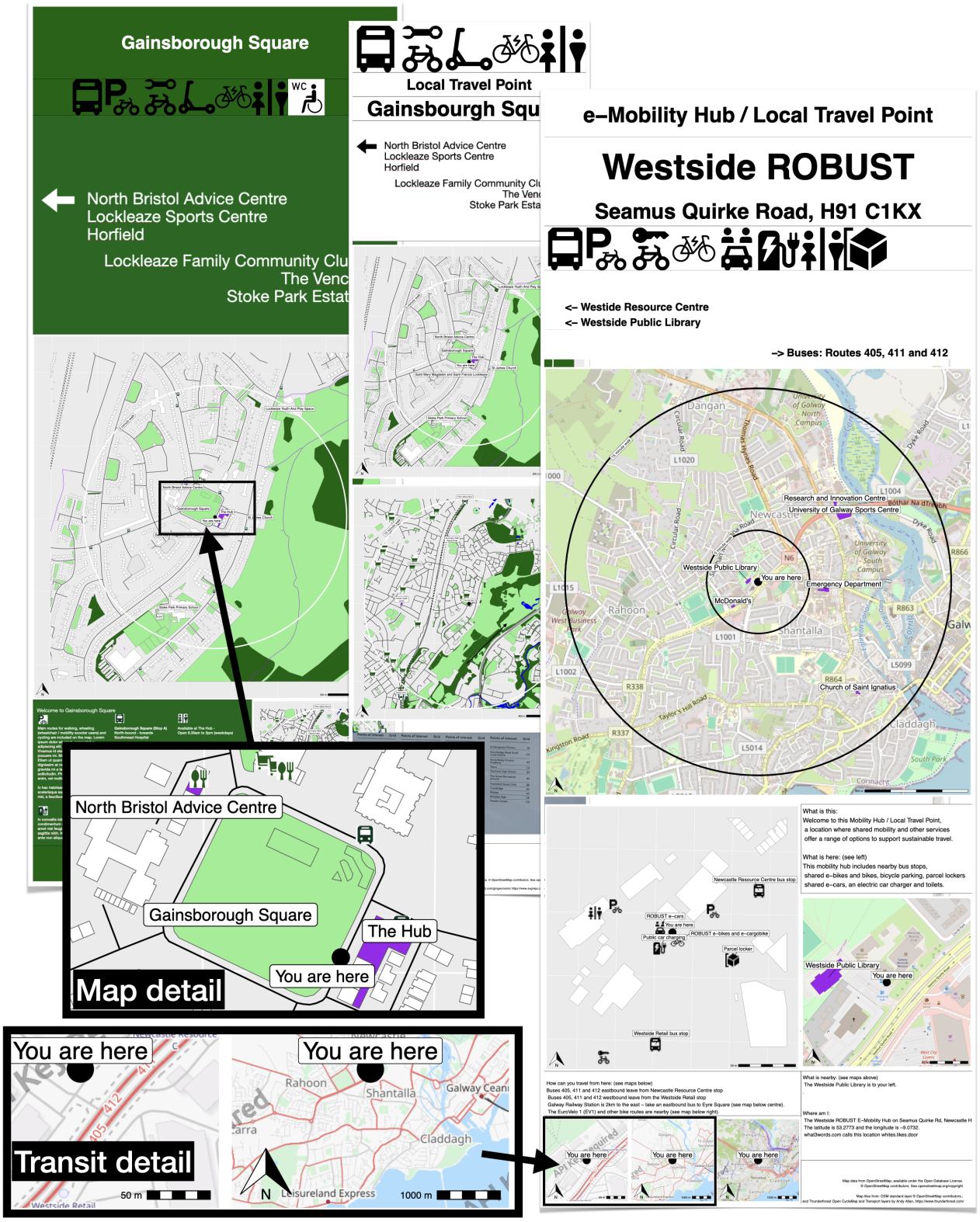

Figure 5: Software output examples: Bristol, Birmingham and synthesis-style totem displays (left, centre, right). Source: authors.